



OFFLINE-FIRST SECURITY ENGINEERING

VNAGY Technical Overview

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1. System Identity

VNAGY is an **offline-first security system** designed for behavior-based detection, modular signal processing, and autonomous protection without cloud dependency. The system uses its own **M-Module mathematical model**, **VNAGY Killing Chain**, **Supply-Chain Security module**, and **VNAGY Rule Format v1.0** as a standardized method for defining security rules. VNAGY is an architectural security layer intended for integration into enterprise environments without degrading existing systems.

VNAGY

2. VNAGY System Overview

VNAGY is a modular, locally executed security system:

- **Offline-first event bus** — all events are processed locally.
- **Local cache** — short-term signal retention without cloud dependency.
- **Local scoring** — mathematical evaluation of behavior.
- **Local Killing Chain** — phased threat detection.
- **Local SIEM exporter** — conceptual output for SIEM systems.

The architecture is fully offline-first, with zero network reliance and zero cloud communication.

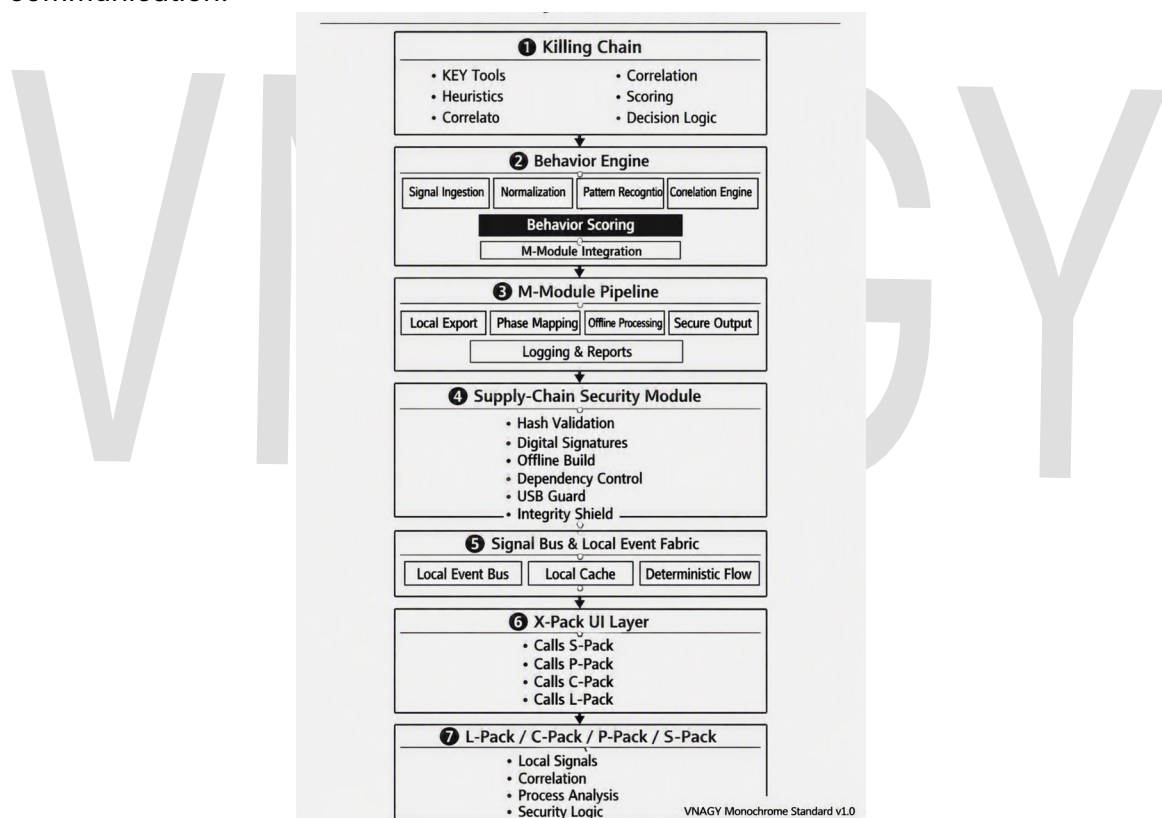


Diagram 1 — VNAGY System Overview

*This diagram shows the complete modular structure of the VNAGY system, including the Killing Chain, Behavior Engine, M-Module pipeline, supply-chain security components, local event fabric, and signal packs. It represents the full architectural layout at a conceptual level. **Generated using AI tools for conceptual visualization. The diagram illustrates architecture only and does not represent operational implementation.***

3. M-Module Mathematical Model (Public Version)

The M-Module is the mathematical core of VNAGY:

- **Input signals** — behavior, processes, network, filesystem, system.
- **Scoring concept** — mathematical evaluation without thresholds.
- **Heuristic concept** — conceptual heuristics without operational detail.
- **Offline-first computation** — local processing without cloud dependency.

Operational elements (thresholds, weights, anti-evasion logic) remain strictly internal.

3.1. Mathematical Framework of the M-Module (Public Version)

by Viktorija Nađ, univ. mag. educ. phys. et comp. Sci. (2026)

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$$M(x) = \sigma(\alpha S' + \beta K + \gamma B) \tag{1}$$

Probabilistic Behavior Analysis Model

Model Components

- **Heuristics (H)** — Memory, CPU, Subnetting
- **Scoring Signals (S)** — Quantitative Indicators
- **Killing Chain (K)** — Attack Phase Correlation
- **Behavior Signals (B)** — Processes, Network, Anomalies

4. VNAGY Killing Chain (MITRE-Aligned)

The **VNAGY** Killing Chain is a phased threat-detection model:

- **Recon**
- **Initial Access**
- **Execution**
- **Persistence**
- **Defense Evasion**
- **Impact**

Each phase is mapped to **MITRE ATT&CK** categories. The model is public, conceptual, and non-operational.

-

5. Supply-Chain Security Module (Concept)

VNAGY includes a dedicated supply-chain security layer:

- **Hash verification**
- **Digital signatures**
- **Dependency freeze**
- **Integrity shield**
- **Anti-corruption guard**

This is a conceptual overview without operational implementation details.

6. VNAGY Rule Format (v1.0)

A standardized format for defining VNAGY security rules:

- **Rule header**
- **Signal definition**
- **Behavior conditions**
- **Killing Chain mapping**
- **Expected outcome**

The format is public but does not reveal operational logic.

6.1. Example Rule — VNAGY Rule Format v1.0 (Simplified Public Version)

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Rule Header

- **Rule ID:** R-001
- **Version:** 1.0
- **Module:** M-Module
- **Category:** Behavior
- **Killing Chain Phase:** Execution
- **Model:** LocalDeterministic
- **Output:** score_value

Correlation Engine

Stage 1: Signal Aggregation

- Process Behavior: detected
- CPU Behavior: detected
- Memory Pattern: detected
- Network Pattern: optional

- Subnet Heuristic: optional

Stage 2: Heuristic Fusion

- H1: active
- H2: active

Stage 3: Behavior Decision

- **Status:** Suspicious
- **Phase:** Execution

Export

- **Format:** VNAGY-LOCAL
- **Target:** Local SIEM

VNAGY

7. Test Concept (Non-Operational)

VNAGY uses conceptual test cases to demonstrate architectural stability and model consistency.

7.1. Behavior Anomaly Test

Purpose: Detect conceptual deviations in process behavior. **Input:** unusual system calls, unexpected file access, atypical network behavior. **Processing:** M-Module evaluates deviation using heuristics and scoring (no thresholds). **Outcome:** informational anomaly signal; Killing Chain does not activate.

7.2. Supply-Chain Integrity Test

Purpose: Detect conceptual integrity violations. **Input:** modified hash, missing signature, version mismatch. **Processing:** hash verification, signature validation, dependency freeze concept. **Outcome:** integrity alert; Killing Chain does not activate.

7.3. Killing Chain Phase Test

Purpose: Validate conceptual phase mapping. **Input:** signals corresponding to each Killing Chain phase. **Processing:** M-Module evaluates; heuristics determine phase; conceptual mapping. **Outcome:** phase-mapped event; no operational output.

8. System Identity Summary

VNAGY is an architectural offline-first security system with its own mathematical model, phased threat-detection logic, supply-chain protection, and standardized rule format. All operational components remain strictly internal.

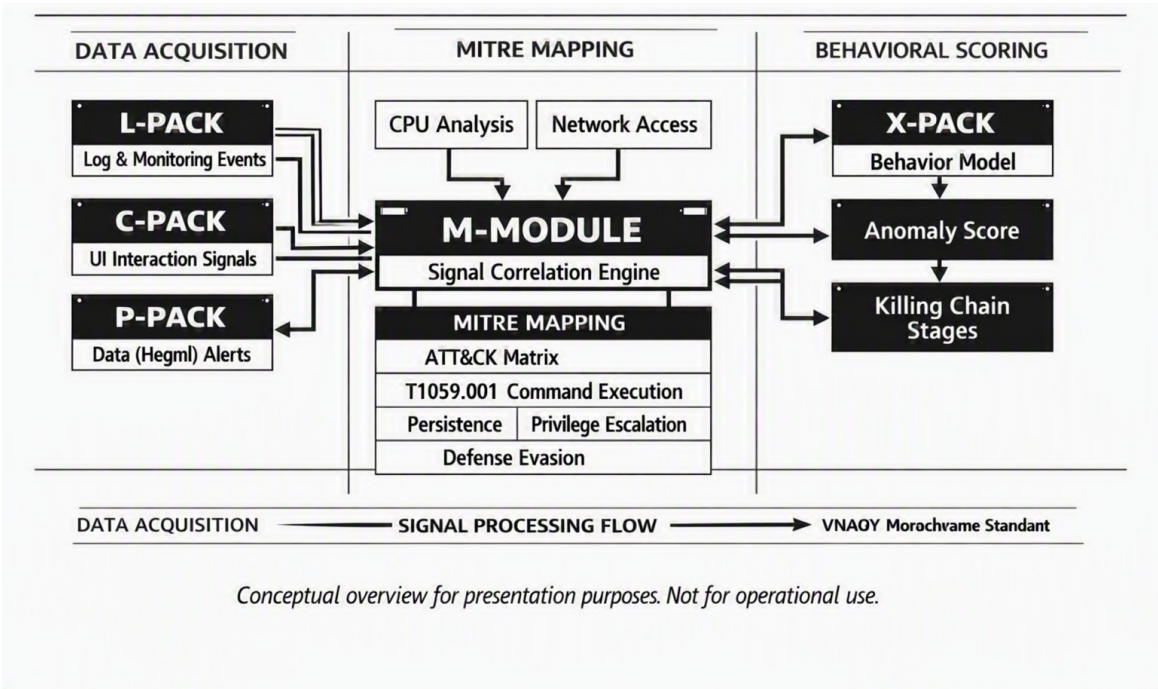


Diagram 2 — VNAGY System Overview

(kratki opis) This diagram shows the complete modular structure of the VNAGY system, including the Killing Chain, Behavior Engine, M-Module pipeline, supply-chain security components, local event fabric, and signal packs. **Diagram generated using AI tools for conceptual visualization. Not for operational use.**

9. Intellectual Property & Licensing

9.1. Ownership

The **VNAGY Security Architecture — CRA Alignment Layer (Preparation Draft)**, including all symbolic constructs, interaction rules, structural definitions, and conceptual models, is the exclusive intellectual property of the author.

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9.2. Permitted Use

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- implementation of any construct, rule, or model
- derivation of operational logic from symbolic definitions
- reconstruction of algorithms, thresholds, weights, or detection semantics
- commercial use in any form
- modification, removal, or alteration of licensing terms
- redistribution of modified versions

9.4. Non-Operational Declaration

This document contains no operational details, no thresholds, no weights, no heuristics, no correlation logic, and no algorithmic semantics. All constructs are symbolic and conceptual. The model is non-computational and non-reconstructable by design.

9.5. License

VNAGY Extended License (PSDL-1.3)

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License reference:

<https://github.com/VNAGY-Framework/VNAGY/blob/main/LICENSE>

VNAGY